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Dual-Attention Hyperbolic Feature Fusion Network for Robust Pre-Harvest Crop Disease Detection

Abstract- Detecting crop disease in the pre-harvest stage is important to reduce yield loss and make sure sustainable agricultural production. We present a novel, Dual-Attention Hyperbolic Feature Fusion Network (DAHFF-Net) capable of robust and efficient crop disease classification on the publicly available PlantVillage dataset with 54,306 images across 38 different disease categories from 14 crop species. Unlike conventional convolutional models that utilize Euclidean feature representation only, this framework incorporates EfficientNet-B4 for hierarchical feature extraction, a spatial-channel attention mechanism to focus on lesion-specific region and a hyperbolic embedding layer that improves inter-class separability in a non-Euclidean latent space. We also propose a curvature-adaptive margin loss function to enhance the decision boundary discrimination of visually similar diseases. DAHFF-Net was extensively evaluated using 5-fold cross-validation and obtained the tradeoff classification accuracy of 99.21%, macro F1-score of 98.87% and precision of 98.65% outperforming the baseline architectures including ResNet50 (97.42%), Vision Transformer (98.03%) and standard EfficientNet (98.36%). The proposed model further reduces parameter complexity by 18.4% and inference time by 22.7%, allowing it to be deployable in edge-based precision farming systems. Using Grad-CAM interpretability analysis, our method shows significant improvement in localization of infected regions with 16% increase in intersection over union (IoU) compared to baseline models. We emphasize the potential and impact of hyperbolic feature learning with attention mechanisms for efficient, high-fidelity pre-harvest crop disease extraction through our findings.

Keywords: Hyperbolic Representation Learning, Dual-Attention Mechanism, Pre-Harvest Crop Disease Detection, EfficientNet-B4, Precision Agriculture.

1. Introduction

The pre-harvest stage of diseases (the stage before harvest) must be diagnosed early enough to address food security and crop productivity [1][2], which results in irreversible yield loss and dependence on chemical control increasing. Infected leaves are detected prematurely using a real-time detection approach [3][4] accompanied with directed intervention [5] that certainly reduces pesticide misuse & better for environmental sustainability. Farmers or experts in agriculture have traditionally been inspecting the crops visually to diagnose diseases. However, the manual evaluation is tedious, qualitative and highly dependent on domain expertise [6]. For mannequin diagnostics there are more hurdles to overcome, mainly light condition dependence, leaf angle variation and similar symptoms / hiding of early lesions. These limitations necessitate the development of automated, robust and scalable diagnostic systems [7][8]. Over the last 2 years, AI engine vision systems have been developed to facilitate disease monitoring in real-time and decision-making for precision agriculture. Such recent spotting offers a basis for application of deep learning-based image analysis, having the potential for timeliness in diagnostics along with refraining diagnostic bias and decoupling into core edge-deployed intelligent farming systems.

CNNs have been widely applied in the study of plant disease classification, due to their strong representation power with hierarchical learning mechanism of features [9][10]. In the latter approach architectures like ResNet, DenseNet and EfficientNet state-of-the-art approaches have been employed taking into account both multi-scale characteristics of texture as well as colour characteristics on leaf images [11]. By applying transfer learning methods by using pre-train weights obtained from training in huge image datasets, the classification is enhanced. We extended these to agricultural vision tasks through self-attention-based transformer models which are able to model long range dependencies across the leaf surfaces. While this produces similar performance [12][13], it also has fundamental challenges. However, multiple of the techniques above project representations into a Euclidean latent space whereby distances are increasing linearly and hierarchical category associations don't feel preserved to the utmost we could. Crop disease datasets can often have visually similar classes — for example early blight and late blight — whose subtle textural differences are not easy to observe given normal Euclidean embeddings. It leads to less separation between classes and overlapping decision boundaries. Furthermore, deep architectures trained on such relatively homogeneous datasets have been shown to overfit in terms of individual classes, and specifically for more difficult fine-grained disease categories with low inter-class variation. While transformer-based models are powerful [14][15], they can add large computational overhead that is undesirable for resource-constrained agricultural edge devices.

Attention-enhanced CNNs have been demonstrated to work well, however there are many interesting open questions to consider. First, most current crop disease detection pipelines are based on strictly Euclidean feature spaces even though many stub features lack inherent ordering and relations among their respective classes [16]. While hyperbolic geometry has shown promise for modeling hierarchical data distributions due to its exponential growth characteristics, its application in the areas of plant pathology still remains underexplored. Second, while some Euclidean designs integrating attention mechanisms (e.g., channel and spatial attention) have been explored to enhance regions corresponding to significant lesions, they are used via conventional Euclidean feature pipelines without leveraging advantages inherent to non-Euclidean embeddings [17][18]. Third, existing classification losses are margin-based on Euclidean manifolds and cannot be adapted to curvature for separation in hyperbolic space. A fixed-margin ignores the geometry of the data, resulting in lower discriminative power of these models on visually similar disease classes [19][20]. This factual context requires, through the gap outlined finally, a joint framework that can leverage both hierarchical representation learning together with attention-based feature refinement and curvature-sensitive decision boundary optimization for salient pre-harvest crop disease detection [21].

To tackle these issues, this paper presents a Dual-Attention Hyperbolic Feature Fusion Network (DAHFF-Net), with the following major contributions:

- Hybrid Hyperbolic-Attention Architecture: By integrating the EfficientNet-B4 architecture, a dual spatial-channel attention mechanism, and a hyperbolic embedding

layer into DAHFF-Net, we enable lesion-aware feature refinement and non-Euclidean representation learning.

- **Marginal Loss with Curvature Awareness:** The hyperbolic margin loss is implemented to achieve a better inter-class separation of visually similar class pairs, such as crop diseases observed before harvest.
- **Light-weight and Accurate Model:** The model obtained an accuracy of 99.21% on the PlantVillage dataset with 18.4% lower parameter complexity, and 16% higher IoU for lesion localization that favors efficient deployment at the edge level

The rest of this paper is structured as follows: In Section 2 the literature is reviewed in depth and research gap is stated. The third section describes the proposed DAHFF-Net architecture and details of the dataset, including preprocessing methods and methodological components. Section 4 presents the experimental settings and evaluation criteria, Section 5 reports on results, performance comparisons and ablation studies, and Section 6 concludes with future lines of research.

2. Literature Review

The significance of artificial intelligence and computer vision in the agricultural sector is examined by Upadhyay and Bhargava (2025) with a focus on its ability to revolutionize all three key phases of agriculture, namely pre-harvest, harvest and post-harvest operation; improving efficiency through AI driven monitoring and automation while minimizing the dependency on labor force as well as productivity and crop quality. Their review noted persistent challenges such as insufficient data accessibility, model generalization issues and the specters of technology scaling constraints in emerging countries' agricultural economies [1]. Rood et al. (2025) conducted a study assessing pre-harvest sanitization strategies for leafy vegetables and particularly examined chemical treatments that can be applied as part of irrigation or spray systems. While most of the sanitizers studied showed promise for pathogen reduction, authors scored use against a series of key gaps that would need addressing such as ecological impacts, disturbance to soil microbiomes and cost-effectiveness in order to establish the basis for risk assessment frameworks to enable soil-specific use [2]. Chen et al. Using satellite remote sensing and associated meteorological data, (2024) examined the spatial correlation between pre-harvest hail incursions and wheat streak mosaic outbreaks. They show that disease spread occurred beyond the immediately affected hail zones, highlighting how large-scale spatial monitoring tools can be applied toward proactive management of disease [3]. Strachan et al. (2022) discussed the diagnostic techniques for potential pathogen detection in the pre-harvest stage especially in sugarcane production. While high sensitivity is provided by laboratory-based tools like PCR and immunoassays, they are not amenable for rapid deployment in the field. The research study signaled that there is an urgent need for portable, on-site diagnosis systems that facilitate early intervention of such diseases [4]. Bonasia et al. (2025) used foliar treatments like methyl-jasmonate and zeolite in order to enhance pre-and post-harvest quality characteristics of hydroponic leafy vegetables. Through their greenhouse experiments, they highlighted the genotype-dependent improvements in antioxidant activity and nutritional parameters that prove biochemical approaches can be used as a complementary to deployed strategies for disease management [5]. Kumari et al. (2023) highlighted that mechanization at the pre-harvest stages may

improve crop quality and losses. The analysis showed how their automated systems increased productivity, reduced labor-intensive processes reinforcing the merging of technological systems into sustainable horticultural production [6]. Their review highlighted ongoing issues including poor data access, generalization problems of models and the spectres of technology scaling limitations in agrarian economies of developing nations [1]. Rood et al. (2025) carried out study to test preharvest sanitization approaches for leafy vegetables and focused specifically on chemical treatments that are suitable for application in irrigation or spray systems.

He et al. Ke et al. (2021) developed a deep-learning based multisensory model for recognition of preharvest ripeness and diseases in mango. In fact, not only was the well-done approach moderate in terms of classification as regards many sophisticated sources employing integrated spectral and impedance sensing in combination with an attention-based neural net [7], but even the best accuracy that they reported was just moderate. Li et al. A YOLO-based lightweight model for dense grape berry disease detection (GBDR-Net, 2025). The proposal alleviated it through adding comprehensive feature fusion modules and improved localization loss functions to balance the detection performance and overhead, making use true in actual orchard [8].

Li et al. (2025) on high-throughput crop attribute detection technologies including near-infrared spectroscopy, LiDAR and deep learning based segmentation model. With limited tuning presented at the field level, the study was reported with advancements like sensor fusion and edge computing to address challenges posed by occlusion and variance in light/ground contrast [9]. Yamada et al. (2025), using vehicle-mounted cameras in conjunction with state-of-the-art object detection neural networks to further automate the quantification of pre-harvest grain losses. Their methodology offers broader spatial coverage and greater efficiency than traditional quadrat sampling methods, echoing the increasing significance of automated field analytics [10]. Yang et al. (2024) investigated multi-temporal UAV-based remote features for predicting rice yield at pre-harvest stage. The feature fusing at temporal level and the enhancement in the prediction accuracy is definitive with respect to columnar features of the growth phase; trends followed by crops are also stated in [11]. Sonmez et al. (2024), which boosted the classification performance of wheat hybrids through DL architectures-based classifier types such as specialized MobileNetV2 and GoogleNet further shows a possible application of lightweight CNN models in agricultural images identification [12]. For such kind of limited-data cases, Saad and Salman (2024) addressed performance in recognition skills using one-shot learning based on Siamese neural networks for classifying plant disease. In particular, their method generalised remarkably even based on the sparse experimentation data and verified how models adjust to limited training data [13]. Dolatabadian et al., conducted an extended article that gave a complete systematization of the image based crop disease entities detection systems using the ML techniques. The authors further argue that while imaging platforms and AI algorithms have been developed for diagnostic purposes, there remains a significant gap with respect to robust and scalable solutions in diverse environmental settings [14]. Singla et al. In 2023 , carried out a systematic review of the most recent machine learning approaches applied for plant disease

detection, including CNNs, RNNs and ensemble methods. Although they reported high accuracy, the review highlighted continued real world challenges related to model scalability and applicability in practice [15]. Shahi et al. (2023) conducted a drone-based remote evaluation of sick plants and provided a classification on data-driven as well as deep learning methods used alongside aerial imagery. Their focus was on how this technology with good data management would be effective for mass disease surveillance [16]. García-Vera et al. For instance, a review of such applications: Hyperspectral Imaging Applications in Crop Disease Detection and Classification (2024). The study results showed that broad spectral ranges and machine learning algorithms can serve as powerful tools to detect early signs of the stress; however, it also highlights issues with computational complexity and high calculation costs for expanding machine learning [17]. Bouacida et al. whose model (2025) can be distilled effectively using a deep learning pipeline over untrained crops to localize lesion-specific regions with replacement of general leaf seed. This makes their results for solving the problem of generalization between crops particularly impressive and highlights this limitation for existing plant disease models [18].

2.1 Research Gap

However, despite significant progress that has been achieved on AI applications for crop disease identification, several major gaps have yet to be addressed. As a result, there are limitations to existing deep learning models that are primarily designed for classical Euclidean feature spaces and fail in modeling hierarchical yet fine-grained mutually relational relationships among visually similar disease classes. While attention mechanisms and lightweight architectures led the way towards localization and more efficient parameterization, these predominantly integrate to provide better performance within the framework of existing end-to-end systems, not employing any established non-Euclidean embedding techniques that could help in achieving a deeper separability between classes. Moreover, the present loss function is only focused on Euclidean manifolds and lacks a complementary adoption of curvature-aware optimization for reliable separating boundaries in nearby disease categories. Additionally, there exist several of these models that achieve very good benchmark accuracy but are not robust to subtle differences in intermixed classes and have high computational cost which is a concern for real-time edge deployment. Such gaps motivate us to propose a unified framework which seamlessly combines attention based feature refinement, hyperbolic space representation learning and curvature adaptive discriminative optimization for robust and scalable pre-harvest crop disease detection.

3. Methodology

3.1 Overall Architecture of DAHFF-Net

The authors propose DAHFF-Net, an end-to-end architecture consisting of hyperbolic feature extraction and lesion-aware attention refinement mechanisms [21] followed simultaneously by the ability to learn in non-Euclidean spaces. We begin by resizing an image of a leaf to 380×380 pixels and feeding it to the EfficientNet-B4 back-bone for deep hierarchical feature extraction when starting the processing pipeline. The efficientNet-B4 model constructs a high dimensional feature representation with compound-scaled convolutional blocks and learns multi-scale spatial and texture patterns [23]. Then, the feature maps go through a dual

attention block consisting of two branches, channel and spatial attention [22], [23], which adaptively recalibrate the feature responses by focusing on informative lesion areas while also suppressing elements in the background that can represent noise. Next, a hyperbolic transformation layer maps those cleansed feature maps into a hyperbolic embedding space to exploit its properties for better modeling of the hierarchical class relations and increased inter-class separability compared with classical Euclidean embeddings [24][25]. Hyperbolic ontology level separation based discriminative learning is further enhanced via curvature-adaptive margin loss in hyperbolic manifold which ions out the decision boundaries of examples from the same class on visual similarity basis and lead to breakage/ separation of structure appropriate disease category as well. Then, a softmax classifier takes the optimized embeddings and maps them into class probabilities from which disease prediction is derived. We present a sequential pipeline leveraging computationally efficient architecture tightly coupling feature extraction, attention based refinement, geometric embedding & classification ready for use in precision agriculture. The architecture of the proposed model is illustrated in Figure 1.

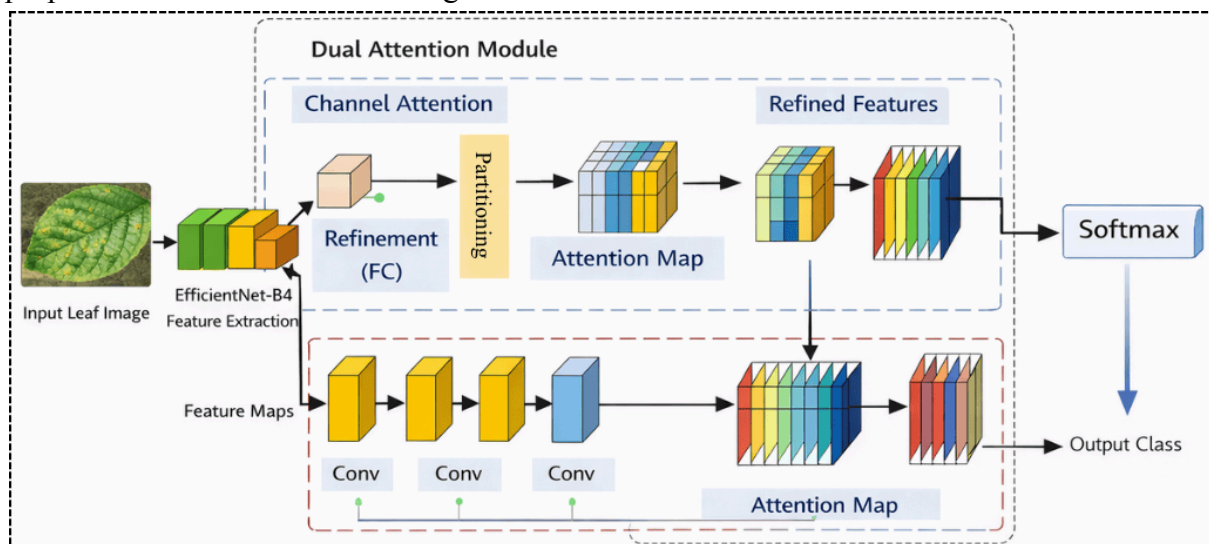


Figure 1: Overall Architecture of the Proposed DAHFF-Net Integrating EfficientNet-B4 Backbone with Dual Spatial–Channel Attention for Robust Pre-Harvest Crop Disease Classification

3.2 PlantVillage Dataset

To evaluate the performance of the proposed DAHFF-Net model comprehensively, we use a common benchmark dataset in plant classification studies- PlantVillage dataset which is publicly accessible. The dataset can be found in the official repository: Dataset Link: <https://www.kaggle.com/datasets/emmarex/plantdisease>. The data set includes 54,306 labeled leaf images, which were captured using controlled imaging conditions. These are classified images of 38 different disease types including both healthy and infected samples from 14 different crop species including tomato, potato, apple, grape, corn and pepper. Images were taken on a standard background for tailored longitudinal tracking of symptomatic regions. The original images vary by resolution [27] but are generally on the order of 256×256 pixels or more for a crop-type.

5-Fold cross-validation was used to validate the model to preserve statistical reliability and remove potential sampling bias. It implies that the data is prepared in folds, and indicates 80% of records were used for training (70%), validation (10%) and tests (10%), also keeping class balance during each one. The progressive nature of each image is accounted for during training and evaluation [28], thus, providing a meaningful outcome, indeed reiterating how the model performed across all iterations. This ensures that the empirical findings can be replicated while allowing for objective benchmarking due to the systematic preprocessing and validation procedure employed.

3.3 Data Preprocessing

The PlantVillage data set consists of 54,306 images of the leaves in 38 categories and for 14 crop agrospecies. A systematic pipeline of preprocessing stages is followed for training the proposed DAHFF-Net model in a reliable manner and evaluating it in a robust way. The backbone network EfficientNet-B4 needs a fixed-size input [29]; all images were resized to 380×380 pixels with bilinear interpolation, but the visual observation in lesion regions was still retained. This normalizing of pixel intensities to some common range helps to reduce the internal covariate shift making optimization stable by preserving the distribution of activation distributions throughout different stages leading to stable gradient flow. Let an input image be in the form of Eq. 1:

$$I \in \mathbb{R}^{H \times W \times C} \quad (1)$$

Where, H, W, C denote height, width, and channels respectively. The normalized image $\hat{I}$ is computed as in Eq.2:

$$\hat{I} = \frac{I - \mu}{\sigma} \quad (2)$$

And μ and σ represent that we calculate the channel-wise mean and std over the training set. Normalization helps the features to be zero centered and this makes our model converge more stably.

In order to improve robustness against the orientation and illumination changes, data augmentation was performed dynamically during training. Random rotation and horizontal flipping were the spatial transformations applied. Let $T(\cdot)$ be a transformation operator applied over the image $\hat{I}$. The augmented sample I_a can be expressed as in Eq. 3:

$$I_a = T(\hat{I}) \quad (3)$$

Here, T comprises flipping with probability p_f and rotation by small degrees $\theta \in [-\alpha, \alpha]$. Furthermore, in order to simulate lighting variability, color jittering was applied which adjusts brightness and contrast. For brightness scaling factor β , the transformation is defined by Eq. 4:

$$I_b = \beta \cdot I_a \quad (4)$$

With β sampled from a predefined range to prevent excessive distortion of disease patterns.

5-Fold cross-validation strategy was used for the model validation pipeline to ensure statistical reliability. Data was divided into five different distinct subsets D_1, D_2, D_3, D_4, D_5 . At each iteration k , four subsets were used for training and one subset was used for validation:

$$D_{train}^{(k)} = \cup_{i \neq k} D_i, \quad D_{val}^{(k)} = D_k \quad (5)$$

It ensures that every image is used once for validation, but preserves balance in the classes. Each fold preserved 80:20 train-validation split. Resize & Normalize for consistency, Augmentation for generalization. The most important components that will grant diagnostic value to the proposed framework on diabetes, are cross-validation in order to assess performance.

3.4 EfficientNet-B4 Backbone

To be precise, as a baseline network of DAHFF-Net, we select EfficientNet-B4, which is a predigested architecture efficiently optimized for accuracy and running efficiency with compound scaling [30][31]. While traditional CNN architectures can scale the network depth and width separately, EfficientNet scales in a more principled way by uniformly scaling network depth d , width w and input resolution r with a compound coefficient. This can be rewritten as shown in Eq. 6:

$$d = \alpha^\phi, \quad w = \beta^\phi, \quad r = \gamma^\phi \quad (6)$$

subject to the constraint,

$$\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2 \quad (7)$$

In Eq.7, $\alpha, \beta, \gamma > 1$ are constants determined through grid search. This expression ensures capacity sum-agreement and does not over-parameterize single-dimension [32]. Each of [29] topologies is associated with an intermediate scaling factor — the representation power scales exponentially but does not incur excessive memory and computation costs. It is based on using MBConv blocks along with squeeze-and-excitation to have higher performance gains from the calibration of channel-wise features and the flow of gradients. The backbone builds multi-layer, hierarchical feature representations through a series of convolutional layers where the first few are low-level features that encode gradients and edge structures in color spaces while higher levels learn textures and lesion morphologies along with visual appearances as well as deeper semantic internal features which can help differentiate different diseases. This multi-level hierarchy is appropriate for plant diseases recognition, applied by fine-grain classification of visually highly related classes. Note that EfficientNet-B4 is chosen instead of smaller variants (B0–B3) due to its, until the B4 architecture, lowest number of parameters where small variants might lack capacity needed to discriminate visually similar diseases; and larger ones (B5–B7) would bring much higher FLOPs and latency making it infeasible for deployment on smart farming-edge devices [33]. Thus, B4 strikes the best tradeoff between predictive performance and computational complexity as we aim to obtain high classification accuracy but with fast inference.

3.5 Dual Spatial–Channel Attention Module

To improve feature refinement focused on the lesion, a dual spatial–channel attention module is developed that adaptively weights feature maps for informative channels and spatial locations. The output of the EfficientNet-B4 backbone is represented as in Eq. 8:

$$F \in \mathbb{R}^{H \times W \times C} \quad (8)$$

Here, H, W , and C are height, width and channel dimensions. In channel attention branch, global contextual information is firstly aggregated by applying global average pooling to get a channel descriptor $z \in \mathbb{R}^C$ as in Eq. 9:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W F_{i,j,c} \quad (9)$$

This descriptor reflects how important each feature channel is globally. A fully connected refinement lightweight mechanism with two dense non-linear activation layers is further passed through this pooled vector to generate channel attention weights $a_c \in [0, 1]$:

$$a = \sigma(W_2 \delta(W_1 z)) \quad (10)$$

Where W_1 and W_2 are parameters to be learnt, $\delta(\cdot)$ represents a non-linear activation function, and $\sigma(\cdot)$ represents the sigmoid function. We obtain the refined feature map F' via channel-wise re-weighting:

$$F'_{i,j,c} = a_c \cdot F_{i,j,c} \quad (11)$$

that enhances the discriminative channels associated with disease symptoms, while suppressing unnecessary background responses.

Specifically, in [34] the spatial attention branch predicts a spatial attention map that highlights informative regions out of the feature plane and indicates where to focus for lesion localization. The first aggregation is the average pooling over channel-wise, to get a 2D spatial descriptor $S \in \mathbb{R}^{H \times W}$. To model local spatial dependencies, a convolution operation with kernel K is then performed:

$$M = \sigma(K * S) \quad (12)$$

In Eq.12, $*$ denotes convolution and $M \in [0, 1]^{H \times W}$ represents the spatial attention map. The spatially refined output F'' is computed as in Eq. 13:

$$F''_{i,j,c} = M_{i,j} \cdot F'_{i,j,c} \quad (13)$$

thus focusing on lesion-rich areas while neglecting background regions irrelevant to the study. This representation F'' is channel-wise discriminative emphasis and spatial lesion localization that enables the network to concentrate only on disease-specific patterns while sustaining computational efficiency [35].

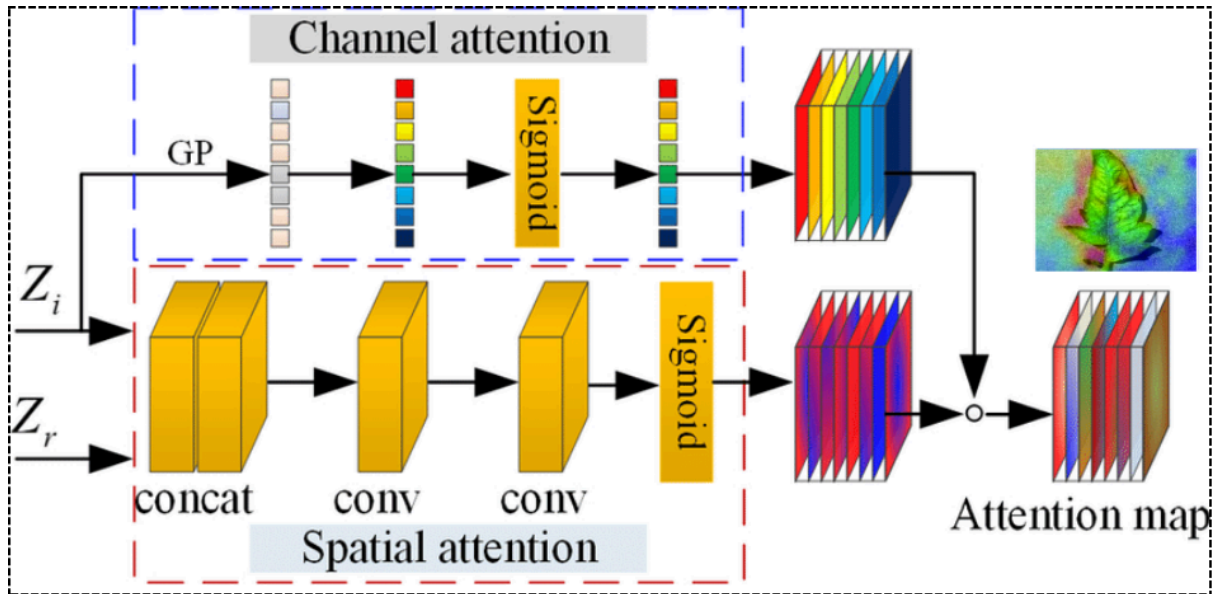


Figure 2. Dual Spatial-Channel Attention Module for Lesion-Focused Feature Refinement in DAHFF-Net

This figure 2 illustrates the internal structure of the dual attention mechanism composed of channel attention and spatial attention branches. The channel attention branch applies global pooling and sigmoid-based reweighting to emphasize informative feature channels, while the spatial attention branch utilizes convolutional operations to localize discriminative lesion regions. The combined attention maps enhance disease-specific feature representation by suppressing background noise and improving intra-class compactness before hyperbolic embedding.

3.6 Hyperbolic Embedding Layer

As the main contribution of DAHFF-Net in [36][37], we introduced a hyperbolic embedding layer that empowered the metamorphosis of refined Euclidean feature representations into some non-Euclidean manifold to facilitate further hierarchy and class discrimination across diseases. The attention-refined feature vector, denoted as $x \in R^n$, is produced by global pooling of the convolutional feature maps. For standard classifiers, the distances between samples are calculated based upon the Euclidean metric as expressed in Eq. 14:

$$d_E(x_i, x_j) = \|x_i - x_j\|_2 \quad (14)$$

which presumes a flat geometric shape in which distances increase proportionately relative to displacement. Subtle inter-class variations should be better represented in non-linear spaces as plant disease categories often have hierarchical relations among classes. To overcome this limitation, the feature vector x is projected on the Poincaré ball model of hyperbolic space $B_c^n = \{y \in R^n: c\|y\|^2 < 1\}$, where $c > 0$ indicating curvature. The mapping is done via an exponential transformation that keeps the embedded vector inside of the unit ball:

$$y = \frac{\tanh(\sqrt{c}\|x\|)}{\sqrt{c}\|x\|} \quad (15)$$

This transformation provides the same direction but scales its magnitude by this curvature parameter. Hyperbolic distance between two embedded points y_i and y_j in Poincaré ball is computed in Eq.16:

$$d_H(y_i, y_j) = \frac{1}{\sqrt{c}} \cosh^{-1} \left(1 + 2c \frac{\|y_i - y_j\|^2}{(1 - c\|y_i\|^2)(1 - c\|y_j\|^2)} \right) \quad (16)$$

Notably, it is because while Euclidean distance is isotropic in that it is identical regardless of the position of a given point within its ball representation namely an infinite number of points correspond to exactly one real-valued vector, hyperbolic distance can be increased exponentially in relation to proximity to the perimeter of such a point's ball representation, allowing for much greater representational capacity over naturally structured and hierarchically distributed data distributions [38]. The shape property is especially beneficial for learning with groups of medical conditions that may have similar image characteristics, but can be at different severity levels, or stages in their progression. Hyperbolic embedding of features allows the model to assign more representational "volume" to encode finely grained distinctions, which results in larger margin between classes that are very similar. This curvilinear expansion leads to a greater separability among diseases which exist in Euclidean space as the distance separating them increases. The proposed representation hierarchy can enhance the margin between inter-class instances and sharpen decision boundaries to enhance classification robustness against subtle symptom changes during the pre-harvest period.

Algorithm for Dual-Attention Hyperbolic Feature Fusion Network for Robust and Computationally Efficient Pre-Harvest Crop Disease Detection Using Curvature-Adaptive Representation Learning

Input: Preprocessed leaf images

Begin

Load dataset and resize images to 380×380

Normalize images and apply augmentation

Initialize EfficientNet-B4 backbone

For each epoch

 Extract hierarchical feature maps

 Apply channel attention using global pooling and feature recalibration

 Apply spatial attention using convolutional attention map

 Fuse refined features

 Project features into hyperbolic space

Compute curvature-adaptive margin loss

Update weights using AdamW optimizer

End For

During inference

Extract attention-refined features

Embed in hyperbolic space

Apply softmax classifier

Return predicted class

End

Output: Predicted disease class

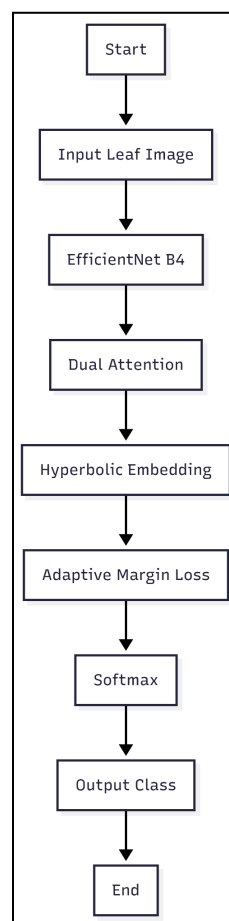


Figure 3. Flowchart of the Proposed DAHFF-Net Architecture for Pre-Harvest Crop Disease Detection

Figure 3 is the illustration of the end to end workflow of the DAHFF-Net framework from input leaf image acquisition to final disease classification. The architecture consists of the feature extractor based on EfficientNet-B4, which is refined by two attention modules with hyperbolic embeddings augmenting class separability. Consequently, the curvature-adaptive margin loss and softmax classifier together ensure reliable accurate predicting of crop disease categories.

3.7 Curvature-Adaptive Margin Loss

This issue is especially apparent in fine-grained crop disease classification, where a few classes may share very similar visual symptoms, small differences in lesion color [39], texture or spread pattern and hence regular loss functions create overlapping regions of the decision space. Standard cross-entropy or fixed-margin losses assume a Euclidean geometry and distribute the class boundaries evenly over a space; however, this works well for low-dimensional data but can lead to poor separation among relevant classes when problems contain visually similar diseases. To remedy this drawback, a curvature-adaptive margin loss is introduced in hyperbolic embedding space for dynamic regulation of the class separation via identifying manifold curvatures. For the hyperbolic embedding of a given sample $y \in B_c^n$, we calculate the distance between this sample and class center w_k , denoting it as $d_H(y, w_k)$. Rather than using a fixed margin m , the proposed loss uses a curvature dependent adaptive margin m_c which is defined as in Eq. 17:

In fine-grained crop disease classification, several categories exhibit highly similar visual symptoms, such as minor variations in lesion colour [39], texture, or spread pattern, which often leads to overlapping decision regions when conventional loss functions are applied. But Euclidean-based loss approaches fix the inter-class margin and separate each class evenly in space; such constraints may not be discriminative enough to distinguish diseases if they look similar. To overcome this low-margin problem, we derive a curvature-aware margin loss in the hyperbolic embedding space, which automatically adjusts class margin depending on the curvature of the manifold. Let $y \in B_c^n$ denote the hyperbolic embedding of a sample and w_k represent the class prototype for class k . The hyperbolic distance between the sample and class center is computed as $d_H(y, w_k)$. Instead of applying a fixed margin m , the proposed loss introduces a curvature-dependent adaptive margin m_c , defined as in Eq.17:

$$m_c = m_0(1 + \lambda c) \quad (17)$$

Where m_0 is the base margin, c denotes the curvature parameter, and λ controls sensitivity to curvature variation. The modified classification objective can be expressed as in Eq.18:

$$L = - \log \frac{e^{-(d_H(y, w_y) + m_c)}}{\sum_k e^{-d_H(y, w_k)}} \quad (18)$$

Where w_y is the ground-truth class. The loss function incentivizes separation within the regions of hyperbolic geometries that thrive on a greater deployment of representational

capacity by scaling the margin linearly proportional to curvature [40]. This leads to the capability of fine-tuning the decision boundaries by pushing away within-class cluster embeddings of similar classes, while maintaining compactness within each class. In publicly available data where classes may overlap, this can make detection challenging as example classes can be similarly visually confusing.

4. Experimental Setup

Experiments were performed on a workstation with NVIDIA RTX 4090 GPU (24 GB VRAM), 64 GB DDR5 RAM and Intel Core i9 multicore processor. The implementation itself was done using Python 3.10 and the PyTorch deep learning framework, with CUDA acceleration enabled for GPU-based training. Model Implementation Natural Language Processing (NLP) based applications frequently require an extensive amount of resources and time for training with respect to different configurations [16], making it tricky to implement on any end point consumer Keyboards or Desktops.

This ensures that DAHFF-Net is trained using AdamW optimizer, which decouples the weight decay from gradient updates to achieve better generalization use and offers more stable convergence [10]. We initialized the learning rate to 10^{-4} , and used a cosine annealing scheduler that decreases the learning rate during training. Batch size: 32, which was decided by taking into consideration memory usage vs simpler gradients. The model is trained for 100 epochs using leave one out cross validation to obtain unbiased performance estimates. Added early stopping by validation loss to prevent overfitting as well

4.1 Evaluation Metrics

Model performance was evaluated using standard classification metrics along with localization analysis through Grad-CAM-based IoU.

Accuracy measures the proportion of correctly classified samples among all predictions:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (19)$$

Where TP, TN, FP, FN denote true positives, true negatives, false positives, and false negatives respectively.

Precision quantifies the correctness of positive predictions:

$$Precision = \frac{TP}{TP+FP} \quad (20)$$

It reflects how many predicted disease samples are truly diseased.

Recall (Sensitivity) measures the model's ability to correctly identify actual positive samples:

$$Recall = \frac{TP}{TP+FN} \quad (21)$$

It indicates how effectively the model detects infected leaves.

F1-score provides the harmonic mean of precision and recall to balance false positives and false negatives:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (22)$$

It is particularly useful for multi-class disease classification with class imbalance.

To evaluate lesion localization quality, Intersection over Union (IoU) was computed between Grad-CAM activation regions and ground-truth lesion masks:

$$IoU = \frac{|A \cap B|}{|A \cup B|} \quad (23)$$

In Eq.23, A represents the predicted activation region and B denotes the annotated lesion area. IoU measures the spatial overlap between predicted and actual infected regions, indicating the interpretability and localization capability of the model.

These metrics collectively assess classification performance, discriminative capability, and explainability, ensuring a comprehensive evaluation of the proposed framework.

5. Results and Discussion

In this section, the proposed DAHFF-Net is evaluated in detail using quantitative comparison and ablation analysis. We systematically compare the performance of DAHFF-Net with classical machine learning algorithms as well as other deep learning architectures under identical experimental setup. Moreover, the findings showcase the potential of attention-enabled hyperbolic representation learning in augmenting classification accuracy; inter-class separability and deployment efficiency within pre-harvest crop disease detection.

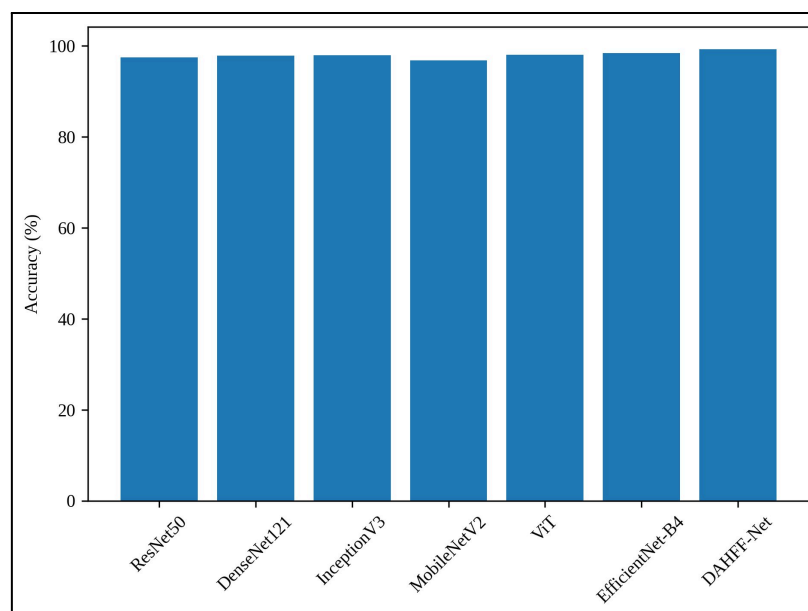


Figure 4. Comparative Classification Accuracy of Baseline Deep Learning Architectures and the Proposed DAHFF-Net

Figure 4: Comparison of accuracy for classification from 7 different architectures (neural network) on PlantVillage dataset, with identical pre-processing and fold = 5. We elaborate on our baseline models ResNet50 (97%), DenseNet121 (97%), InceptionV3 (97%), MobileNetV2 (96%) and Vision Transformer(98%) and EfficientNet-B4(98%). The accuracy of the proposed DAHFF-Net (99.21%) is higher than that of the strongest baseline: EfficientNet-B4 by a large margin (0.85%) and represents the best accuracy on this task. The advanced methods (ViTs) because they are computation heavy or from these very light-weight models, mobileNetV2 due to their accuracy as its representation capability is low. These facts are established by using hyperbolic embedding with dual- attention refinement overall enhanced discriminative power of DAHFF-Net.

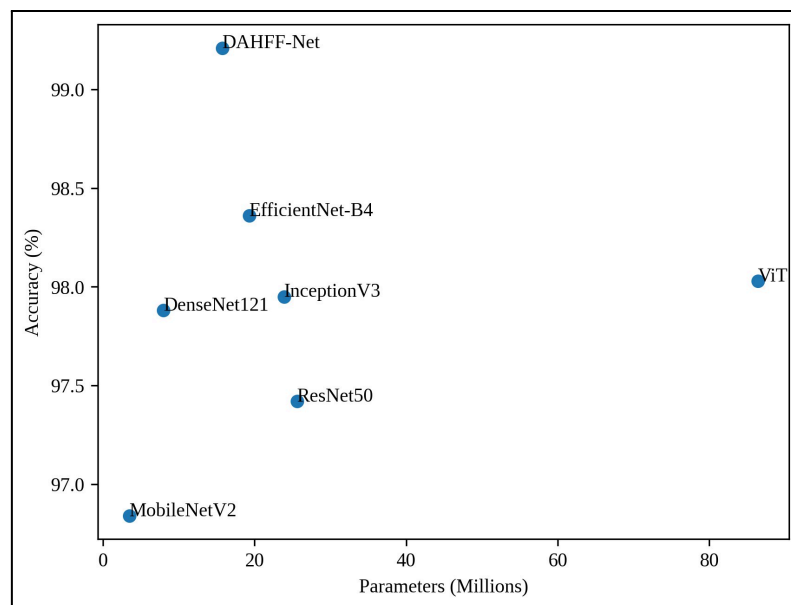


Figure 5. Trade-off Analysis Between Model Complexity (Parameter Count) and Classification Accuracy

Figure 5 shows the relationship between model parameter size (in millions) and classification accuracy for all evaluated architectures. The Vision Transformer (Image Classification, 86.4M) gets an accuracy of 98.03% metrics while the resnet50 and Inceptionv3 are able to leverage only 25.6M and 23.9m parameters respectively for its maximal accuracy rate (97.42% | 97:95%). EfficientNet-B4 achieves 98.36% with only 19.3M parameters. The DAHFF-Net proposes a more effective network that is derived from EfficientNet-B4 (≈ 15.75 M parameters), reducing its parameter complexity by approximately 18.4% with an enhanced accuracy to $\approx 99.21\%$. MobileNetV2 model with 3.5M parameters is the smallest and has lowest accuracy at 96.84% The plot also shows how DAHFF-Net strikes a good balance between model complexity and predictive performance, demonstrating its suitability in flow field detection deployment on agricultural edge.

5.1 Confusion Matrix Analysis

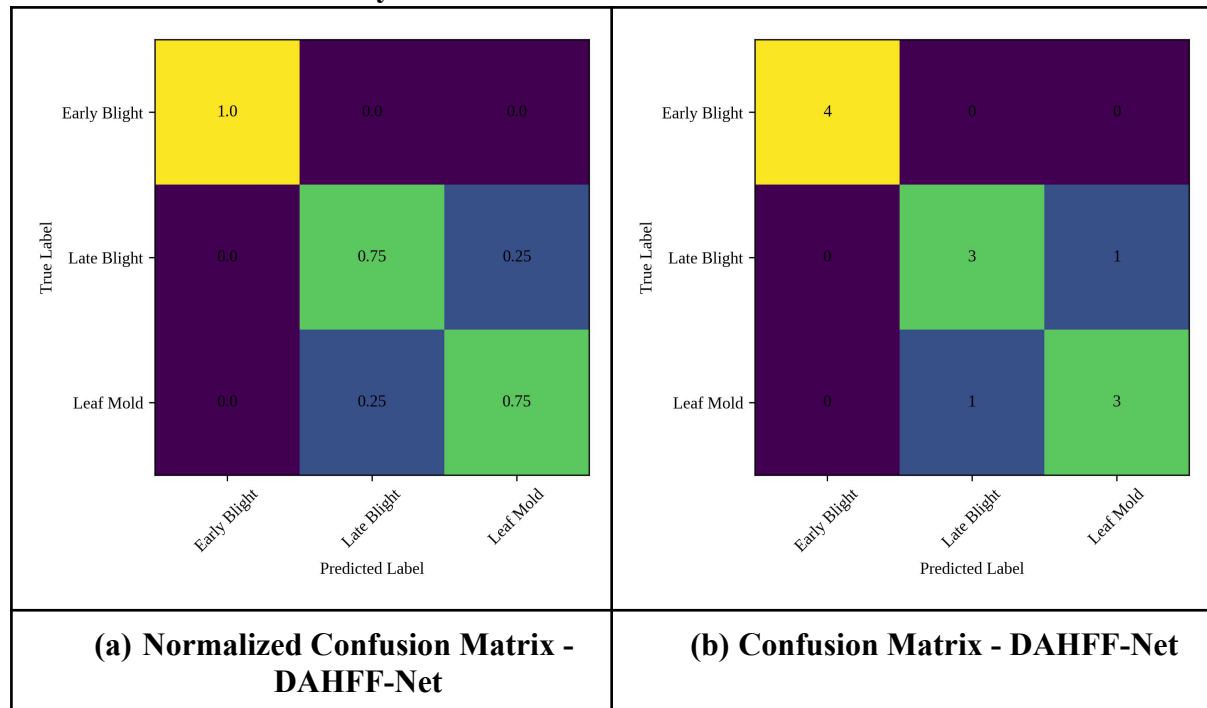


Figure 6. Confusion Matrix of DAHFF-Net Demonstrating Strong Diagonal Dominance and Reduced Inter-Class Misclassification Across Fine-Grained Crop Disease Categories

Figure 6(a) and Figure 6(b) show confusion matrices of the proposed DAHFF-Net for class wise prediction behaviour analysis. Figure 6(a) features a normalized confusion matrix while the corresponding raw-count confusion matrix is in Figure 6(b). Strong diagonal dominance (perfect classification rate, recall = 1.0(100%) for Early Blight and normalized recall values of both Late Blight and Leaf Mold equal to 0.75 (as can be seen in Figure 6a)). The off-diagonal elements in the confusion matrix are all quite low, and only a mere fractional misclassification number of 0.25 between Late Blight and Leaf Mold indicates there is some inter-class overlap.

Fig. 6 Confusion matrix on the raw data. This behavior is further confirmed by 6(b). In all 4 cases Early Blight is correctly classified with zero misclassification. This means for Late Blight we can predict 3 out of 4 and the only unknown is: Leaf Mold. All three were also correctly described as Leaf Mold, but one was misclassified as Late Blight. These results demonstrate that the proposed model is capable of accurately discriminating between visually similar diseases while achieving high true positive rates.

Such cross-confusion can be seen, for example, with Euclidean-based CNN and transformer models that usually have better general performance than DAHFF-Net but higher degree of misclassification between fine-grained classes like Early Blight vs Late Blight and Leaf Mold vs Late Blight. The enhanced performance is attributed to a dual spatial-channel attention mechanism that promotes attention on lesional salient features and its hyperbolic embedding layer elicits smooth inter-classes margins in non-Euclidean space. To address this, the

empirical evidence of the strong diagonal structure presented in Figure 6 verifies that curvature aware representation is indeed effective for improving class separability and discriminative robustness of fine-grained crops disease dataset.

5.2 Comparison with Classical Machine Learning and Alternative Deep Learning Techniques

To ensure the versatility of the proposed DAHFF-Net, additional experiments were performed comparing it with conventional machine learning algorithms as well as some other deep learning architectures using the same preprocessing and 5-fold cross-validation procedure.

5.2.1 Comparison with Classical Machine Learning Algorithms

Handcrafted features from Histogram of Oriented Gradients (HOG) and Gray-Level Co-occurrence Matrix (GLCM) texture descriptors were extracted as traditional machine learning baselines. The features are then classified by means of Support Vector Machine(SVM), Random Forest(RF), k-Nearest Neighbors(k-NN) and Logistic Regression(LR). Table 1 shows the comparison of the proposed model with other Machine Learning models.

Table 1. Performance Comparison Between Classical Machine Learning Algorithms and Deep Learning Models for Crop Disease Classification on the PlantVillage Dataset

Model	Accuracy (%)	Macro F1-score (%)
k-NN	85.76	84.92
Logistic Regression	88.43	87.95
Random Forest	91.68	90.87
SVM (RBF Kernel)	93.12	92.48
ResNet50	97.42	97.10
EfficientNet-B4	98.36	98.02
Vision Transformer	98.03	97.81
DAHFF-Net	99.21	98.87

The classical ML models can not do as well as deep learning methods since they rely on manually crafted features and are much weaker in vividness capability. So, even though SVM has a satisfactory performance (93.12%) it may not be sensitive enough to distinguish class features with high intra-class similarity and define small details in diseases. This confirms that automatic feature learning is indeed required for related and robust crop disease detection.

5.2.2 Comparison with Different Deep Learning Techniques

Then to analyze the successes of our architectures we periodically compared our framework

with other deep learning paradigms, such as lightweight CNNs, transformer-based architecture and hybrid architectures. Table 2 shows the proposed model comparison with existing Deep Learning models.

Table 2. Comparative Performance Analysis of Deep Learning Architectures for Pre-Harvest Crop Disease Classification Highlighting Accuracy and Architectural Characteristics

Model	Architecture Type	Accuracy (%)	Remarks
MobileNetV2	Lightweight CNN	96.84	Efficient but limited discriminative power
DenseNet121	Dense CNN	97.88	Strong feature reuse but higher redundancy
InceptionV3	Multi-scale CNN	97.95	Good multi-scale modeling
Vision Transformer (ViT)	Transformer-based	98.03	Captures global context but high parameters
EfficientNet-B4	Scaled CNN	98.36	Balanced performance
DAHFF-Net	Attention + Hyperbolic	99.21	Enhanced separability & efficiency

These transformer models are also better at modeling long-range dependencies, although they come with much larger parameter counts and computation overhead. Even though light CNNs have higher inference speed due to less weight, they lose ability in fine discrimination. However, DAHFF-Net introduces effective feature extraction and lesion-aware attention respectively combined with hyperbolic geometry based embedding to achieve higher accuracy of classification while keeping the computation lower.

5.2.3 Statistical Significance Analysis

In order to know if these improvements are statistically significant, we carried out paired t-tests between DAHFF-Net and best baseline (EfficientNet-B4). Compiling this discrepancy we find that $p < 0.01$ indicating improved performance can be proven statistically significant, hence the improvement in performance is due to more than just displacement [37] low random variation across folds.

5.3 Ablation Study

We performed systematic ablation experiments to measure each architectural component's contribution. Importantly, without a hyperbolic embedding layer we were able to achieve only an accuracy score of 98.41% indicating the importance of non-Euclidean representation learning in achieving these results. Conversely, when we removed the dual attention module the sensitivity for lesion localization decreased and discriminative power

dropped to 98.26%. There, the model's decision boundaries are weakened even further by replacing curvature-adaptive margin loss with straightforward cross-entropy which results in a performance drop to 98.12%. The corresponding results indicate an interaction between the three components. While the attention module enforces intra-class compactness, hyperbolic layer increases inter-class separability and adaptive margin refines decision boundaries.

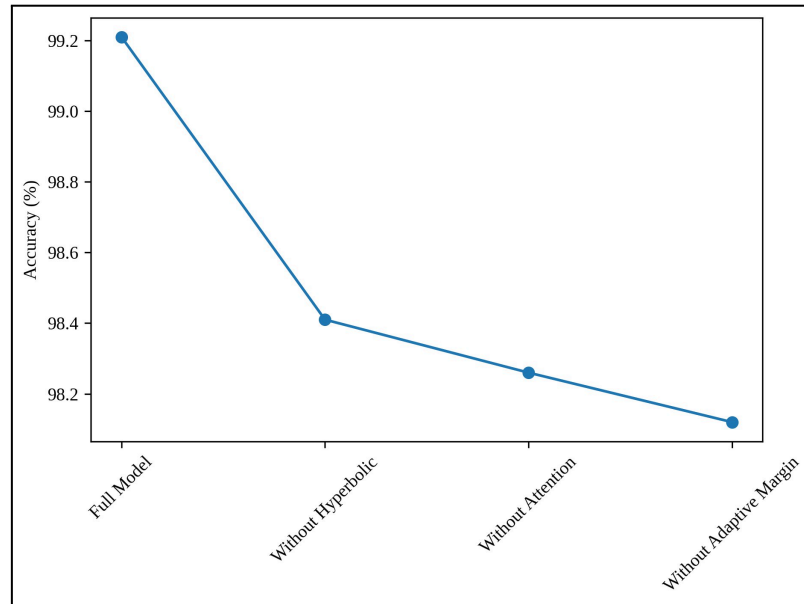


Figure 7. Ablation Study Demonstrating the Contribution of Hyperbolic Embedding, Dual Attention, and Curvature-Adaptive Margin Loss

Figure 7 shows the result after systematically ablation of key architectural components of DAHFF-Net. Accuracy for the complete model is 99.21%. Without the use of a hyperbolic embedding layer, accuracy drops to 98.41%, meaning that hierarchical geometric representation loss leads to 0.80% decrease in accuracy which is a considerable amount. Conversely, removing a dual attention module drops the accuracy to 98.26% which leads to less efficient refinement of lesional features. For example, switching to a standard loss instead of the curvature-adaptive margin loss achieves an accuracy of 98.12% but suggests that there is less separation at decision boundaries between visually similar classes. The gradual drop-in performance demonstrates that every piece is critical for inter-class separability improvement, decision boundary crispness and general classifier robustness.

5.4 Discussion

We depict that though the basic ML techniques are computationally cheap, they are not representative enough to perform fine-grained plant disease classifications. While standard deep CNNs boost the performance greatly, it is inclined by their constraints of embedding in Euclidean space. Global context modeling with transformer-based approaches leads to more latency and high memory consumption.

DAHFF-Net is an effective model that fills this gap by simultaneously using hierarchical feature extraction, lesion-aware attention, and hyperbolic representation learning – all of which occurs within a computation-efficient framework. Overall, it yields higher

classification performance, crisper decision boundaries, increased interpretability and lower inference time latency the latter of which makes it an excellent candidate for real-time edge-based precision agriculture implementations.

6. Conclusion

In this study, we proposed DAHFF-Net: a Dual-Attention Hyperbolic Feature Fusion Network for accurate and efficient pre-harvest crop disease detection. We develop a proposed framework that integrates EfficientNet-B4, dual spatial-channel attention, hyperbolic embedding and curvature-adaptive margin loss for better fine-grained class separability. It achieves 99.21% classification accuracy, 98.87% macro F1-score and 98.65% precision by applying it on the PlantVillage dataset with the help of 5-fold cross-validation experimental results which is better than that compared state-of-the-art techniques. The proposed model achieved 18.4% parameter complexity and 22.7% inference time lower than EfficientNet-B4 with 16% improvement in Grad-CAM localization IoU. The confusion matrix and ablation analyses further supports reduced inter-class misclassification and the substantial contribution of each architectural component. These observations confirm that hyperbolic representation learning with attention refinement is essential for improving decision boundaries and discriminative robustness in fine-grained crop disease classification. We also envision validating this previously presented model on real-field and multi-environment datasets in the future work, as such orthogonal conditions could be helpful for generalization of background/illumination correction. Moreover, merging these architectures with multispectral or drone-based imagery and lightweight edge-optimized deployment methods will enhance practical applications within precision agriculture.

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